LAB REPORT

IBM HR Attrition Analysis using R

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1. Introduction

Employee attrition is a major concern for organizations as it affects productivity, cost, and workforce stability. Data analytics helps in identifying patterns and factors responsible for employee turnover.

In this project, we performed exploratory data analysis and machine learning on the IBM HR Attrition dataset using R. The aim is to analyze key factors influencing attrition and build a predictive model.

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2. Problem Statement

To analyze employee data and:

- Identify factors affecting employee attrition
- Perform data visualization and statistical analysis
- Build a predictive model to classify employees as likely to leave or not
- Provide insights for reducing attrition

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3. Dataset Description

Dataset used: **IBM HR Employee Attrition Dataset**

Features include:

- Age
- Monthly Income
- Job Role
- Department
- Years at Company
- Job Satisfaction
- OverTime
- Attrition (Target Variable)

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4. Tools & Technologies

- R
- ggplot2
- tidyverse
- corrplot
- caret
- randomForest

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5. System Architecture

Dataset → Data Cleaning → Exploratory Data Analysis → Visualization → Feature Engineering → Model Training (Random Forest) → Prediction → Evaluation

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6. Methodology

Step 1: Data Loading

The dataset is loaded into R using read.csv() and inspected using summary functions.

Step 2: Data Cleaning

- Missing values checked
- Constant columns removed
- Target variable converted to factor

Step 3: Exploratory Data Analysis (EDA)

- Overall attrition rate calculated
- Distribution of attrition across categories analyzed
- Numerical features analyzed using boxplots

Step 4: Data Visualization

- Bar charts for categorical variables
- Box plots for numerical variables
- Correlation heatmap for feature relationships

Step 5: Feature Interaction Analysis

Attrition analyzed based on combined factors like:

- Department × OverTime

Step 6: Model Building

A Random Forest model is trained to predict attrition.

Step 7: Model Evaluation

- Confusion matrix generated
- Accuracy and performance evaluated

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7. Algorithm

1. Load dataset
2. Perform data cleaning
3. Convert target variable to factor
4. Perform EDA
5. Visualize categorical and numerical features
6. Compute correlation matrix
7. Split data into training and testing sets
8. Train Random Forest model
9. Predict on test data
10. Evaluate using confusion matrix

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8. Results

Key Observations:

- Employees working overtime have higher attrition
- Lower job satisfaction leads to higher attrition
- Employees with lower income are more likely to leave
- Certain job roles show higher attrition rates

Model Performance:

- Random Forest model provides good classification accuracy
- Important features identified using feature importance

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9. Advantages

- Data-driven decision making

- Identifies key attrition factors
- Visual insights for HR teams
- Predictive modeling capability

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10. Limitations

- Dataset is static (not real-time)
- Model accuracy depends on data quality
- External factors not included

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11. Applications

- HR analytics
- Employee retention strategies
- Workforce planning
- Business decision making

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12. Future Scope

- Use advanced models (XGBoost, Deep Learning)
- Deploy as a dashboard (Shiny App)
- Real-time employee monitoring
- Integration with HR systems

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13. Conclusion

This project successfully analyzes employee attrition using R and machine learning techniques. The insights obtained can help organizations reduce attrition and improve employee retention. The Random Forest model effectively predicts employee turnover and highlights important contributing factors.